BUILDING THE TWIN

Data platforms

Module 5 · Where the twin keeps its memory

Transport Twin Study Path

A typical data path



Streaming or batch?

Streaming

Seconds

Live state

Control and alerts

Batch

Hours to days

History

Calibration and learning

The right store for each job

- Time series: TimescaleDB, InfluxDB
- Geometry and networks: PostGIS
- Live context: NGSI-LD brokers, twin graphs
- Raw archive: object storage, Parquet

Four levels of interoperability

- Technical: can systems connect?
- Syntactic: can they parse the format?
- Semantic: do they mean the same thing?
- Organisational: may they share it?

Design for federation

Own your data, share by standard

Open APIs and shared vocabularies let twins connect later